

1. Understanding AI and Generative AI

Artificial Intelligence (AI) enables computers to analyze data, recognize patterns, solve problems, and support decision-making. Generative AI extends these capabilities by creating new content such as text, images, quizzes, worksheets, lesson plans, and assessments from user prompts.

For a TGT Science teacher, AI simplifies lesson planning, develops competency-based questions, explains scientific concepts, and creates engaging classroom activities. It also supports personalized learning by adapting content to students' needs. However, AI-generated information should always be verified for scientific accuracy and curriculum alignment. Teachers must review and modify content to suit students' learning levels. AI is a valuable teaching assistant, but it cannot replace a teacher's professional judgement, scientific knowledge, and classroom experience.

2. AI and Education

AI is transforming science education by making learning more personalized and interactive. It helps teachers prepare lesson plans, assessments, worksheets, and competency-based questions while reducing administrative work such as grading and attendance. Students benefit from instant explanations, adaptive learning, and additional practice.

Despite these advantages, AI may produce inaccurate, outdated, or biased information. In science, incorrect explanations or experiments can create misconceptions. Therefore, teachers must verify AI-generated content using reliable sources such as NCERT textbooks and the CBSE curriculum. They should also adapt activities to students' abilities and available classroom resources.

As UNESCO (2023) emphasizes, AI should support educators rather than replace them. When combined with critical thinking and professional expertise, AI enhances teaching and improves learning outcomes.

Reference: UNESCO. (2023). *Guidance for Generative AI in Education and Research*.

3. My Experience Using Generative AI as a TGT Science Teacher

As a TGT Science teacher, I used ChatGPT to prepare a lesson plan on "**Exploring Mixtures and Their Separation**." I asked it to generate learning objectives, competency-based questions, classroom activities, simple experiments, worksheets, and assessment ideas. The response was well-organized and significantly reduced the time required for lesson planning. It also suggested creative activities that could make science learning more engaging.

While reviewing the content, I found that some examples were not fully aligned with the NCERT textbook, and a few activities required laboratory resources that were unavailable in my school. Instead of using the material directly, I verified every scientific concept with the NCERT Science textbook, CBSE learning outcomes, and other reliable educational resources. I also modified the activities to suit my students' learning levels and classroom environment.

This experience showed that ChatGPT is an excellent teaching assistant but not a replacement for professional judgement. Teachers should always verify scientific facts, review AI-generated content for accuracy, and ensure it is age-appropriate, inclusive, and curriculum-aligned. When used responsibly, AI enhances science teaching while teachers remain central to meaningful learning.

References: NCERT Science Textbooks (Classes VI–X); CBSE Science Curriculum and Learning Outcomes.